

PHISHING EMAIL DETECTOR USING RULE-BASED ANALYSIS

1. Project Overview

The Phishing Email Detector is a cybersecurity awareness tool designed to analyze email content and identify whether an email is potentially legitimate or a phishing attempt. The system uses predefined rules and keyword analysis to detect suspicious characteristics commonly found in phishing emails. This project is suitable for beginners and helps students understand basic cybersecurity concepts.

2. Problem Statement

Phishing emails are one of the most common cyberattacks used by attackers to steal sensitive information such as usernames, passwords, banking credentials, and personal data. Many users fail to recognize phishing attempts, making them vulnerable to cyber threats. This project aims to provide a simple solution for identifying potentially malicious emails using rule-based detection techniques.

3. Objectives

- Analyze email content for phishing indicators.
- Detects suspicious keywords commonly used in phishing attacks.
- Identify urgency-based phrases intended to create panic.
- Count and analyze links present in the email.
- Calculate a risk score based on predefined rules.
- Classify emails as Safe, Suspicious, or High Risk.
- Improve cybersecurity awareness among users.

4. Features

4.1 Suspicious Keyword Detection

The system scans email content for commonly used phishing keywords such as:

- Urgent
- Verify Account
- Click Here
- Winner
- Prize
- Free
- Password Reset
- Login
- Bank Account
- Confirm Identity

Each detected keyword contributes to the overall risk score.

4.2 Link Analysis

The system identifies URLs present within the email and evaluates them based on quantity and suspicious patterns. Emails containing multiple links are considered more risky.

4.3 Urgency Detection

Many phishing attacks use fear and urgency to manipulate users. The system detects phrases such as:

- Immediate Action Required
- Account Will Be Suspended
- Verify Now
- Respond Within 24 Hours
- Act Now

These phrases increase the phishing risk score.

4.4 Risk Score Calculation

The application assigns points to different phishing indicators. The final score represents the likelihood that an email is a phishing attempt.

4.5 Email Classification

Based on the calculated score, emails are categorized into:

- Safe
- Suspicious
- High Risk

5. System Architecture

User Input Email



Text Processing



Keyword Analysis



Link Detection



Risk Score Calculation



Classification Engine



Result Display

6. Technology Stack

- Python
- Regular Expressions (Regex)
- Tkinter (Optional GUI)
- VS Code

7. Sample Input

Subject: Urgent Account Verification

Dear Customer,

Your account will be suspended within 24 hours.

Please verify your account immediately to avoid service interruption.

Thank you.

8. Sample Output

Detected Keywords:

- Urgent
- Verify
- Suspended
- Immediately

Links Found: 1

Risk Score: 55

Result: HIGH RISK PHISHING EMAIL

9. Advantages

- Easy to understand and implement.
- Introduces basic cybersecurity concepts.
- Helps users identify phishing attempts.
- Suitable for beginners and academic projects.
- Requires minimal resources.

10. Limitations

- Uses predefined rules only.
- Cannot detect advanced phishing techniques.
- Does not verify domain reputation.
- May generate false positives or false negatives.

11. Future Enhancements

- Machine Learning based phishing detection.
- Email attachment scanning.
- URL reputation checking.
- Gmail and Outlook integration.
- Real-time phishing database support.
- PDF report generation.

12. Conclusion

The Phishing Email Detector is a simple yet effective cybersecurity awareness project that helps users identify potentially malicious emails using rule-based analysis. The project demonstrates fundamental concepts of phishing detection and cybersecurity while remaining easy to develop and understand for beginner-level students.